Exercise Sheet 10

Conventions follow Lecture 10. Haar integrals are normalised. On functions, $(\lambda(g)f)(x) = f(g^{-1}x)$ and $(\rho(g)f)(x) = f(xg)$. On $\mathrm{U}(1)$ write $\chi_n(e^{i\theta}) = e^{in\theta}$ and $\int f = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta$. On $\mathrm{SU}(2)$ write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

Exercises 10.1–10.5 are the core. Exercise 10.6 is an optional extension.

Exercise 10.1 (The averaged operator in finite dimensions). Let $K = \mathrm{U}(1)$ act on $H = \mathbb{C}^3$ by $\Pi(z) = \mathrm{diag}(z, z, z^2)$. Let $u = \frac{1}{\sqrt{3}}(1,1,1)^T$ and $E_u v = \langle u, v\rangle u$.

(a) Compute the matrix $T = \int_K \Pi(z) E_u \Pi(z)^{-1}\,dz$.
(b) Find the eigenvalues and eigenspaces of T. Check that each eigenspace is invariant.
(c) Now let $K = \mathrm{SU}(2)$ act on $\mathbb{C}^2$ in the standard way, with $u = e_1$. Compute T in two ways.

Exercise 10.2 (Strong but not norm continuity). Let $K = \mathrm{U}(1)$ act on $L^2(K)$ by ρ. Use the Fourier expansion $f = \sum_n c_n \chi_n$ with $\|f\|_2^2 = \sum |c_n|^2$.

(a) Show that $\rho(e^{it})\chi_n = e^{int}\chi_n$.
(b) Show that $\|\rho(e^{it})f - f\|_2 \to 0$ as $t \to 0$, for each f.
(c) Show that $\|\rho(e^{it}) - 1\| \geq \sqrt{3}$ for every $t \notin 2\pi\mathbb{Z}$. Conclude that $t \mapsto \rho(e^{it})$ is not continuous in operator norm.

Exercise 10.3 (Evaluation at the identity). Let $K = \mathrm{U}(1)$. For a finite set $S \subset \mathbb{Z}$ put $W_S = \mathrm{span}\{\chi_n : n \in S\}$.

(a) Show that W_S is ρ-invariant. Find $u_W \in W_S$ with $f(1) = \langle u_W, f\rangle$ for $f \in W_S$. Verify $f(x) = \langle u_W, \rho(x)f\rangle$.
(b) Compute $\|u_W\|_2$. Show that $\sup\{|f(1)| : f \in \mathscr{R}(\mathrm{U}(1)),\ \|f\|_2 = 1\} = \infty$.
(c) Show that every finite-dimensional ρ-invariant subspace of $L^2(\mathrm{U}(1))$ equals W_S for some finite S.

Exercise 10.4 (The first coefficient block of SU(2) and the tensor order). Let $G_{ij}(g) = g_{ij}$ for $g \in \mathrm{SU}(2)$.

(a) Show that $\rho(h)G_{ij} = \sum_k h_{kj} G_{ik}$. Deduce that each row span $\mathrm{span}\{G_{i1}, G_{i2}\}$ is ρ-invariant, and that $\rho(h)$ has matrix h in the basis (G_{i1}, G_{i2}).
(b) Show that $\lambda(g)G_{ij} = \sum_k \overline{g_{ki}}\, G_{kj}$. Deduce that $\lambda(g)$ has matrix $\bar{g}$ on each column span.
(c) Find $w \in \mathrm{SU}(2)$ with $\bar{g} = wgw^{-1}$ for all g. Describe E_1 under ρ, under λ, and under $K \times K$.
(d) State the Gram matrix of $\sqrt{2}G_{11}, \sqrt{2}G_{12}, \sqrt{2}G_{21}, \sqrt{2}G_{22}$.

Exercise 10.5 (Integrals from the second coefficient block). Realise V_2 inside $\mathbb{C}^2 \otimes \mathbb{C}^2$ with $e_1^2 = e_1 \otimes e_1$, $e_2^2 = e_2 \otimes e_2$ and $e_1 e_2 = \frac{1}{2}(e_1 \otimes e_2 + e_2 \otimes e_1)$. The group acts by $g \otimes g$.

(a) Show that $g \cdot e_1^2 = a^2 e_1^2 + 2ab\, e_1 e_2 + b^2 e_2^2$.

(b) Show that $e_1^2,\ \sqrt{2} e_1 e_2,\ e_2^2$ is an orthonormal basis. Compute the coefficients c_{u', e_1^2} for u' in this basis.

(c) Deduce $\int |a|^4$, $\int |a|^2 |b|^2$, $\int |b|^4$ and $\int \bar{a}^2 b^2$. Check them against $\int (|a|^2 + |b|^2)^2 = 1$.

Exercise 10.6 (Optional extension: a dimension count). Let P_N be the space of functions on $\mathrm{SU}(2)$ given by complex polynomials of degree at most N in $a, b, \bar{a}, \bar{b}$. Let E_m be the coefficient block of V_m.

(a) Show that $E_m \subset P_m$ and $P_N \subset \bigoplus_{m \le N} E_m$. You may use the Clebsch–Gordan rule: $(\mathbb{C}^2)^{\otimes k}$ is a sum of copies of V_m with $m \le k$.

(b) Deduce $\dim P_N = \sum_{m=0}^{N} (m+1)^2 = \frac{(N+1)(N+2)(2N+3)}{6}$. Tabulate $N = 0, \dots, 4$.

(c) Show that $\binom{N+4}{4} - \binom{N+2}{4}$ gives the same numbers. This number is the dimension of the space of polynomials with complex coefficients of degree at most N in four real variables, taken modulo multiples of $|x|^2 - 1$.